NVDA Risk Analysis Project

Diagnostics-first, reproducible risk analytics for NVIDIA (`NVDA`)

Sample window: `2020-01-01` to `2024-12-31`

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Why This Project Exists

- NVDA is a high-volatility, event-sensitive stock with multiple risk channels.
- A one-off notebook can show point estimates, but it does not show data lineage, fallback behavior, or build reliability.
- This project treats reproducibility and diagnostics as part of the deliverable, not as afterthoughts.

What The Pipeline Produces

Pipeline: `data_management -> analysis -> final -> documents`

- Shared daily and monthly panels
- Five risk modules in one workflow
- Presentation-ready figures and summary tables
- Diagnostics, provenance, and scorecard artifacts

Data Rules And Audit Trail

- Sample window fixed to `2020-01-01` through `2024-12-31`
- Source priority: `cache -> snapshots -> online`
- Every build emits `data_provenance.json`
- Diagnostics record consistency checks, fallback events, and status flags

Risk Coverage At A Glance

- Market risk
- Liquidity risk
- Drawdown risk
- Systematic risk
- Macro risk

One pipeline produces all five views from a shared data backbone.

Selected Findings

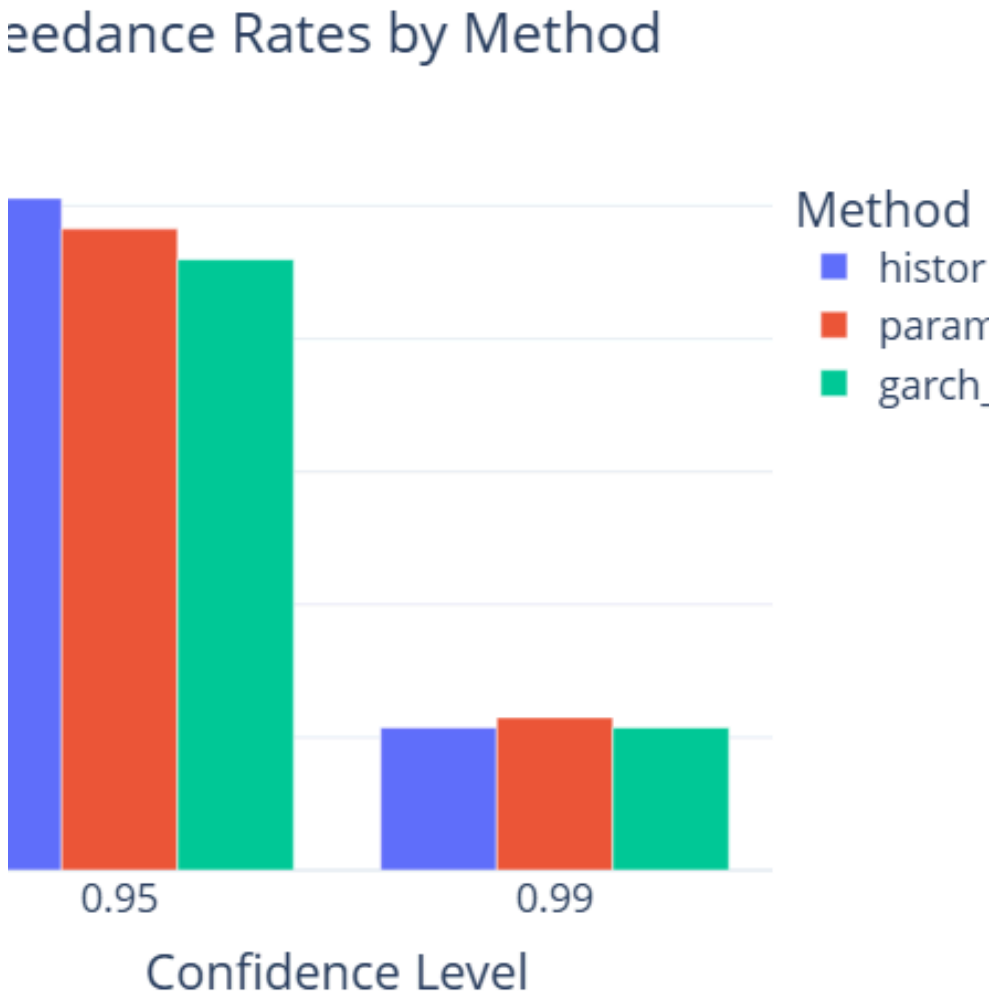
- Annualized volatility: `33.3%`
- Historical VaR (95%): `-13.8%`
- Expected shortfall: `-17.3%`
- Maximum drawdown: `-23.8%`
- Median dollar volume: about `$746.9M`
- Macro DGS10 coefficient: `0.175` with low explanatory power (`R^2 = 0.0007`)

Volatility View

The rolling volatility figure gives regime context and is exported directly into the paper and presentation workflow.

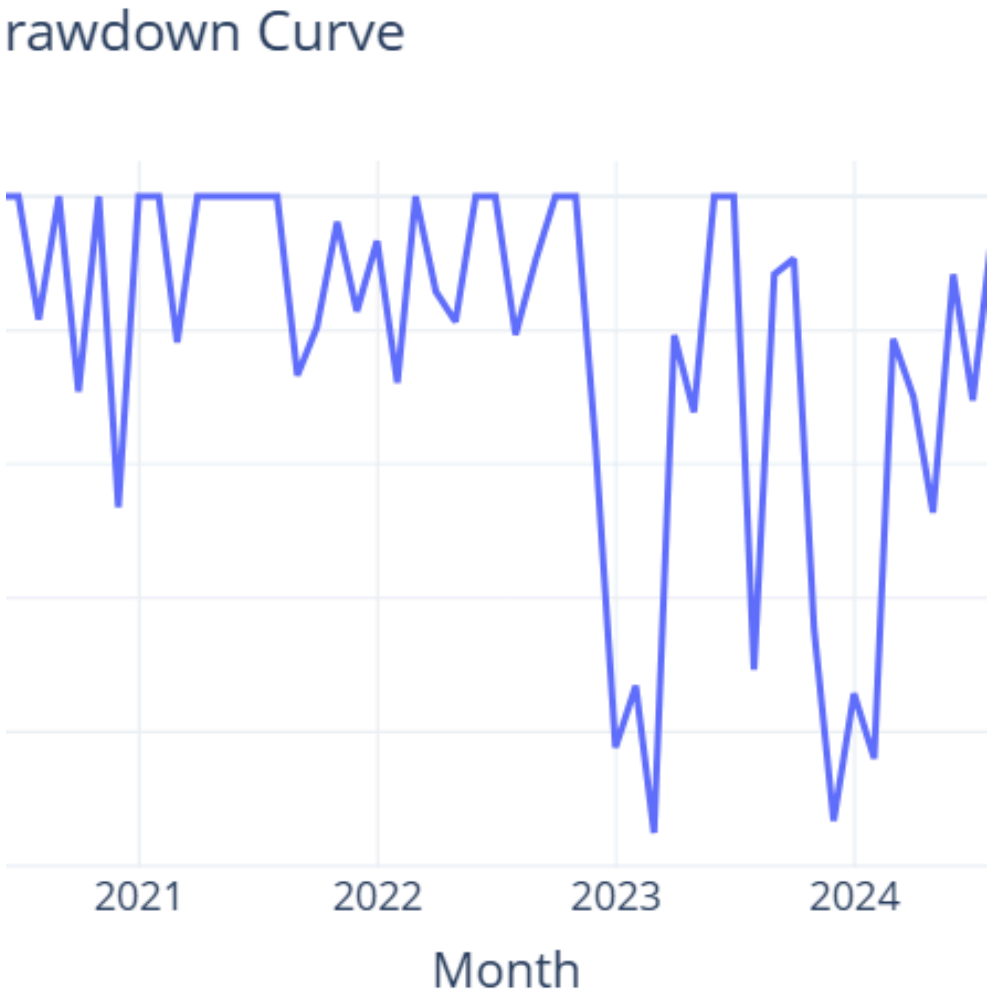
VaR Diagnostics

- Exceedance counts are compared with expected exceedance rates by method.
- Backtests are paired with diagnostics so model failures do not stay silent.



Stress Visualization

- Drawdown figures make both depth and persistence visible.
- Current summary outputs report a maximum drawdown of `-23.8%` .



Credibility Checks

- Latest diagnostics result: `32/32` quality gates passing
- Latest local verification baseline: `133` tests passed
- Coverage baseline: `92.65%`
- Public artifacts include diagnostics, scorecard, and provenance metadata

Reproduce The Project

```
pixi install
pixi run npm install
pixi run pytest -q
pixi run pytask
```

Focused reruns:

```
pixi run pytask -k diagnostics --force
pixi run pytask -k figure --force
```

Thank You

Questions?